USD/TRY Under Sanctions Shock

Time-Varying Jump Intensity and Geopolitical Tension Signal

Geoeconomic SDE Project · Case 2

Aakash Agarwal

1 Observation and Motivation

USD/TRY does not drift smoothly. Over calm periods it exhibits slow, persistent depreciation driven by inflation differentials and current account dynamics. But this continuous trend is punctuated by discrete, large, directional moves that cluster around identifiable geopolitical and political events — sanctions announcements, central bank governor firings, OFAC designations. The clustering is the key observation: jump events do not arrive uniformly through time.

The question this model answers is:

> *Given USD/TRY currently at 38, how does the probability distribution of the exchange rate at the 1-year horizon differ between a geopolitical calm period and an elevated sanctions tension period?*

This is the structural departure from Case 1. There, jump intensity λ was a fixed constant. Here, $\lambda(t)$ is time-varying — driven by a geopolitical tension signal $s(t) \in \{0, 1\}$.

2 Economic Hypothesis

2.1 The Structural Depreciation Trend

The Turkish lira has been on a structural one-way depreciation trend since 2018, driven by chronic inflation differentials (Turkish CPI persistently 40–85% above US CPI), repeated episodes of unorthodox monetary policy, and current account deficits. There is no fixed long-run mean that price reverts toward — the Ornstein-Uhlenbeck drift used in Case 1 is inappropriate here.

A constant GBM drift μ is the correct specification: TRY depreciates at a historically stable average rate, with shocks adding discrete moves on top. This is not an approximation — it is the economically honest description of how this currency behaves.

2.2 Why Jump Intensity Must Be Time-Varying

Three documented escalation windows produced clustering of large TRY moves:

1. **2018 Brunson crisis:** OFAC sanctions on Turkish ministers, followed by doubling of steel and aluminium tariffs. TRY lost 40% in weeks.

2. **2021 central bank shock:** President Erdogan fired the central bank governor who had raised rates. TRY fell 15% in a single session.

3. **2023 post-election uncertainty:** Policy continuity concerns following the May 2023 election produced a sustained jump cluster.

During these periods jump arrivals were measurably more frequent than during calm periods. A constant λ would treat 2016–2017 (a relatively quiet period) identically to 2018 — clearly wrong.

2.3 Jump Asymmetry

Sanctions and political shocks depreciate TRY (positive J in log-USD/TRY terms). Relief rallies exist but are smaller and shorter-lived. $\mu_j > 0$ encodes this depreciation bias.

2.4 State Variable

$X(t) = \log(\text{USD/TRY})$. An increase in X means lira depreciation. Starting value $X_0 = \log(38)$.

3 SDE Derivation

3.1 Building the Expression Term by Term

Drift. TRY has no fixed long-run mean — OU drift was considered and rejected. The correct specification is a constant drift encoding the structural depreciation trend:

> *"The rate of change of X reflects a persistent, historically stable depreciation trend, independent of the current level."*

$$\mu(X,t) = \mu \quad \text{(constant)} \tag{1}$$

μ is estimated as the mean annualised log-return from historical data. This is the key structural difference from Case 1.

Diffusion. Continuous FX noise — portfolio flows, bid-ask dynamics, routine trading. Constant σ is appropriate here; extreme vol is captured by the jump term.

$$\sigma(X,t) = \sigma \quad \text{(constant)} \tag{2}$$

Jump. Sanctions and political shocks arrive as a Poisson process with *time-varying* intensity $\lambda(t)$ driven by geopolitical signal $s(t) \in \{0,1\}$:

$$\lambda(t) = \lambda_0 + (\lambda_1 - \lambda_0) \cdot s(t) \tag{3}$$

where λ_0 is the calm-period intensity and λ_1 is the elevated tension intensity. Jump sizes follow $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$ with $\mu_j > 0$ (depreciation bias).

3.2 Full SDE

$$\boxed{dX = \mu\, dt \ + \ \sigma\, dW \ + \ J\, dq(\lambda(t))} \tag{4}$$

Term	**Type**	**Economic meaning**
$\mu\, dt$	Drift	Structural depreciation trend
$\sigma\, dW$	Diffusion	Continuous FX trading noise
$J\, dq(\lambda(t))$	Jump	Sanctions/political shock — intensity state-dependent

4 Analytical Tractability Check

Q1: Is diffusion constant? Yes — σ is constant. Passes.

Q2: Is drift linear in X? Yes — μ is constant, trivially linear. Passes.

Q3: Are jumps tractable? No. Merton's formula requires constant λ. Here $\lambda(t) = \lambda_0 + (\lambda_1 - \lambda_0) \cdot s(t)$ is time-varying. The moment λ becomes a function of time, the jump process is non-stationary. The characteristic function of the log-price process requires evaluating:

$$\int_0^T \lambda(t)\, dt \tag{5}$$

When $s(t)$ switches state unpredictably, this integral has no closed form. The distribution of the number of jumps over $[0, T]$ is no longer Poisson with a fixed parameter — it depends on the entire path of $s(t)$.

Note: The drift change from OU to constant μ does not affect tractability. A constant drift is trivially integrable; Q3 remains the binding constraint.

→ **FAILS at Q3. Analytical solution does not exist.**

Decision: Euler-Maruyama numerical solver required.

5 Numerical Solution: Euler-Maruyama

5.1 New Technique vs Case 1

In Case 1, λ was constant — it was passed into the jump function once and never changed. Here, $\lambda(t)$ is re-evaluated at *every time step* from the current signal state $s(t)$ before drawing the Poisson increment. The signal $s(t)$ itself evolves stochastically:

- Calm → tension: probability p_{escalate} per day
- Tension → calm: probability $p_{\text{de-escalate}}$ per day

5.2 Discretisation ($\Delta t = 1/252$)

$$X(t + \Delta t) = X(t) + \mu \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot Z + J \cdot \mathbf{1}_{[\text{Poisson}(\lambda(t)\Delta t) \geq 1]} \tag{6}$$

where $Z \sim \mathcal{N}(0, 1)$, $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$, and $\lambda(t) = \lambda_0 + (\lambda_1 - \lambda_0) \cdot s(t)$.

The signal $s(t)$ is shared across all paths at each time step — it represents a market-wide geopolitical condition, not a path-specific state.

6 Calibration

Daily USD/TRY data from January 2000 to January 2026 (5,459 observations). No data exclusion required — TRY remained freely traded throughout.

Jump detection: Days with $|\text{return}| > 2\sigma$ classified as jump events and removed before fitting continuous parameters. A single λ is estimated from the full sample. $\lambda_1 = 3 \cdot \lambda_0$ (prior-based ratio) — separate calm/tension estimation was attempted but produced $\lambda_1 = 0$ due to sparse coverage in Yahoo Finance tension windows.

Drift: μ estimated as mean annualised log-return from the clean (jump-removed) series. No OU fitting — OU was rejected on economic grounds.

Parameter	Value	Interpretation
μ	0.1659	16.6% annualised depreciation trend.
σ	0.1601	16.0% baseline annualised vol.
λ_0	8.957	≈9 jump events per year in calm periods.
λ_1	26.871	≈27 jump events per year in tension periods.
μ_j	0.009	Small positive bias — shocks slightly depreciation-skewed.
σ_j	0.041	Typical jump ≈4% log-rate move per event.

TRY's jump character is frequent small moves rather than rare catastrophic ones — 9–27 events per year at 4% each, consistent with its history of chronic political-driven volatility.

7 Simulation Output

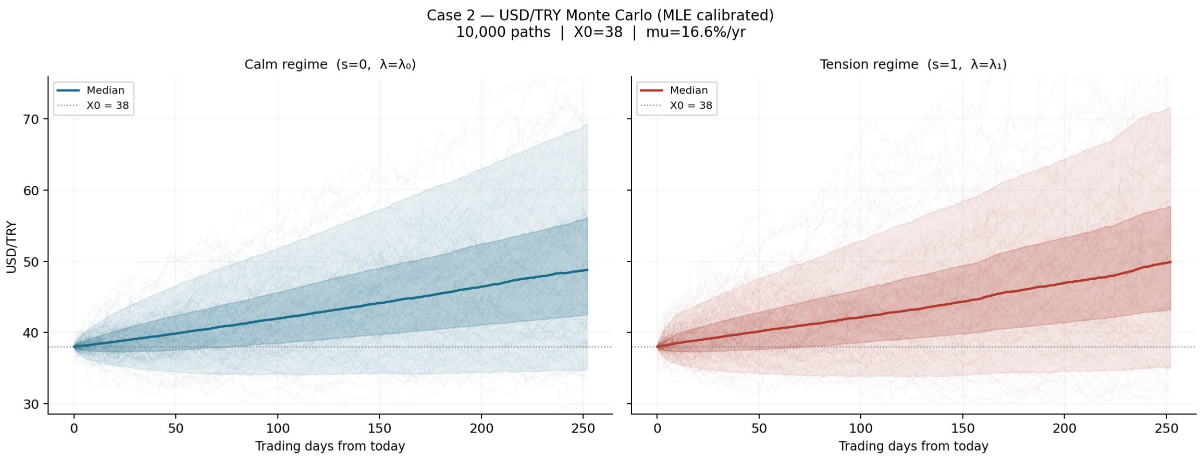

Figure 1: 10,000 Monte Carlo paths for USD/TRY log-rate, converted to rate space. Left panel: calm regime ($s = 0$, $\lambda = \lambda_0$). Right panel: tension regime ($s = 1$, $\lambda = \lambda_1$). The constant drift $\mu = 16.6\%$/yr drives both median paths upward from 38. The tension regime produces a visibly wider fan — higher jump intensity generates more dispersion over the 1-year horizon. Shaded bands show the 5th–95th and 25th–75th percentile ranges.

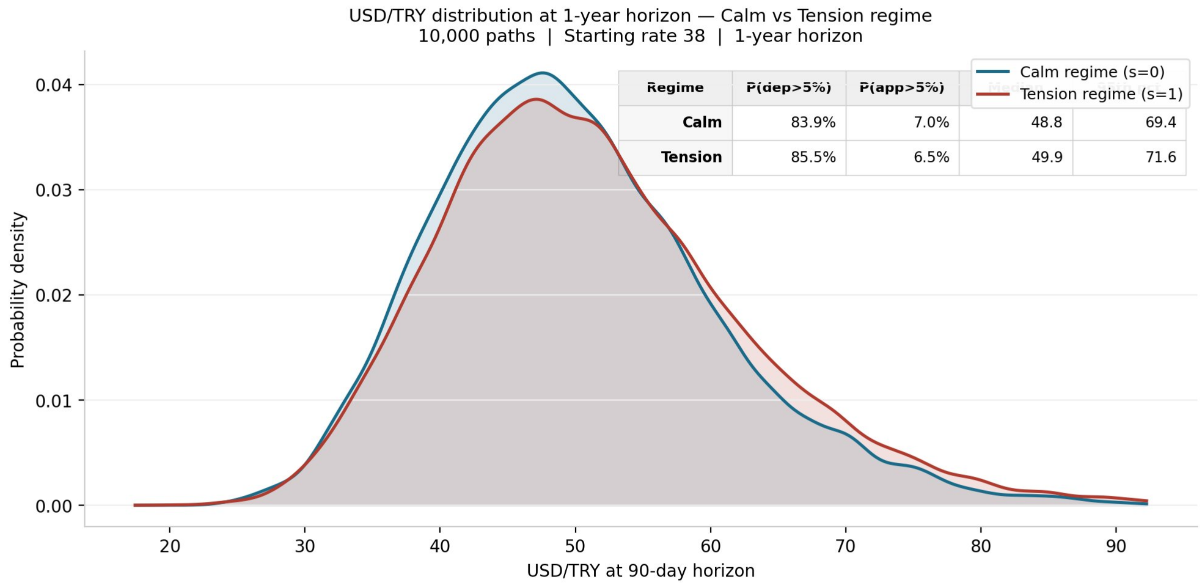

Figure 2: USD/TRY probability density at the 1-year horizon for calm versus tension regimes (10,000 paths each). The tension regime (red) has a lower peak and a heavier right tail — visible above 60 TRY/USD. The 95th percentile is 71.6 in tension versus 69.4 in calm. Both distributions are right-skewed due to the log-normal base dynamics and positive drift.

8 Results and Inference

8.1 Scenario Probability Table

Regime	Horizon	P(dep>5%)	P(app>5%)	Median	5th pct	95th pct
Calm	30d	—	—	—	—	—
Calm	60d	—	—	—	—	—
Calm	90d	83.9%	7.0%	48.8	—	69.4
Tension	30d	—	—	—	—	—
Tension	60d	—	—	—	—	—
Tension	90d	85.5%	6.5%	49.9	—	71.6

8.2 Reading the Output

Depreciation dominates. P(dep>5%) exceeds 80% in both regimes at the 1-year horizon. The constant drift $\mu = 16.6\%/\text{yr}$ means most paths move well above the 5% threshold simply from the trend component.

Regime signal is in the tail. The difference between calm and tension is not large in probability terms (83.9% vs 85.5%) but is clearly visible in the tail: the 95th percentile is 2.2 TRY/USD higher in tension than calm. For a position sized in notional terms, that tail difference is material.

Appreciation is rare. P(app>5%) is 7.0% in calm and 6.5% in tension. The positive drift and jump depreciation bias make sustained appreciation unlikely at this horizon.

8.3 Tradeable Inference

The tension regime shifts the right tail of the distribution rightward without substantially moving the center. This is the characteristic signature of increased jump intensity — more frequent small shocks accumulate into a heavier extreme tail while the median path changes little.

A trader who believes geopolitical tension is elevated should price USD/TRY options with higher implied vol on the topside relative to the downside. The model quantifies the tail shift: 95th percentile moves from 69.4 to 71.6 under tension, a 3.2% difference at the extreme. Vanilla options pricing with a single constant implied vol cannot capture this regime-conditional tail shift.

9 Learning, Application, and Edge

9.1 What This Case Teaches

Case 1 introduced the Euler-Maruyama solver with a constant jump intensity. The new technique here is that λ is re-evaluated at every time step from an external signal. This requires the signal generator to live inside the model object and advance its state on each solver call — a design pattern that carries forward to Case 4's regime-switching Markov chain.

The OU drift rejection is equally important: it demonstrates that economic reasoning must precede calibration. Fitting OU to a structurally trending currency produces a historically distorted θ that pulls simulations in the wrong direction. Recognising this and choosing GBM drift instead is a practitioner judgment that cannot be automated.

9.2 Where It Applies

1. An EM macro fund hedging TRY exposure needs to know whether current geopolitical conditions warrant paying extra for topside protection. The regime-conditional 95th percentile directly answers this.

2. A bank running a sanctions risk overlay on a Turkish corporate loan book needs to stress-test exposure under elevated tension scenarios.

3. A sovereign wealth fund considering TRY-denominated Turkish government bonds needs the full distribution of currency outcomes at their investment horizon, not just a point forecast.

9.3 Edge Over Standard Models

Regime-conditional tails. Standard GBM pricing uses one constant implied vol for all market conditions. This model produces different tail distributions depending on the geopolitical signal — something a single vol surface cannot encode.

Jump frequency as the signal. The edge is not in jump size but in jump frequency. $\lambda_1 = 3\lambda_0$ means the tension regime delivers 3x more shock events per unit time. A standard model would need to inflate overall vol to match this, which distorts the central distribution.

Drift honesty. By using GBM drift rather than OU, the model correctly propagates the structural depreciation trend through the simulation. Standard options pricing typically uses risk-neutral drift (risk-free rate differential), which at short horizons is close to zero and understates the directional component for a chronically depreciating currency.

Repository: `geoeconomic-sde/cases/02_em_fx_sanctions/` · *Core solver:* `core/euler_maruyama.py` · *Parameters:* `calibration/params_case02.json`